

HyperGranular-RAG: Structured Evidence Completion for Compact Multi-Hop Retrieval

Anonymous ACL submission

Abstract

Multi-hop question answering requires complementary evidence within a limited context. A relevance ranking can miss a bridge fact, while indiscriminate expansion can remove evidence that the generator already needs. We study structured evidence completion through HyperGranular-RAG, a method that groups candidate sentences by embedding geometry, selects query-conditioned facet hyperedges, and inserts a bounded number of candidates after a protected dense prefix. In three closed-candidate evaluations on HotpotQA and MuSiQue, the historical compact-MiniLM implementation improves answer F1 by 0.01140–0.01478 with Qwen2.5-1.5B-Instruct; all three recorded paired bootstrap intervals exclude zero. A component comparison favors facet-conditioned selection over a centroid-only selector. The original system nevertheless trails BGE, and neither a cross-space sidecar nor a BGE-native reconstruction establishes an increment over BGE. Protected placement outperforms front insertion in the cross-space study, although this contrast does not isolate position from final context composition. We connect these results to precise definitions of coverage, candidate selection, and evidence displacement, while documenting the historical scoring and statistical limitations. The evidence supports bounded completion for the tested compact backbone, rather than general superiority over strong retrieval.

1 Introduction

Retrieval-augmented question answering separates access to evidence from answer generation. Dense retrieval makes relevance ranking efficient (Karpukhin et al., 2020), but multi-hop questions require several facts to be available together. A passage about one entity may be highly relevant without supplying the bridge needed to connect it to another. HotpotQA and MuSiQue provide complementary settings for examining this difficulty

(Yang et al., 2018; Trivedi et al., 2022).

The challenge becomes sharper under a fixed evidence budget. Adding a candidate does not enlarge a Top-20 context indefinitely: it can promote an already retrieved sentence or displace a different one. Moreover, a generator does not use all retrieved information equally. Evidence position affects context use (Liu et al., 2024), so improved retrieval coverage need not produce an improved answer. A useful completion method must balance candidate relevance, complementarity, and the cost of disturbing the original ranking.

We investigate a structured completion layer, HyperGranular-RAG, that preserves the retrieval backbone. It organizes the available sentence embeddings into local granular balls, uses query-term facets to relate seed balls to candidate expansion balls, and places at most four candidates after ten protected entries. *Completion* describes an intervention on the evidence list; it is not a guarantee that all necessary facts have been recovered. The historical selector is a static, seed-relative heuristic, not a dynamically optimized coverage objective.

Our central question is when this intervention improves end-to-end answers. We examine a compact MiniLM backbone, its component contrasts, transfer to one additional generator configuration, and two extensions around BGE. These comparisons distinguish improving a particular compact retriever from improving a stronger system. They also distinguish choosing a useful candidate set from placing that set less disruptively.

The results reveal a consistent but bounded opportunity. The compact system improves over MiniLM Dense in three evaluations, and its facet-conditioned selector exceeds the specified centroid-only comparator. The original compact system remains below BGE. Neither transferring its candidates to a BGE ranking nor reconstructing the geometry in BGE space establishes additional answer quality over BGE. Thus, the positive compact

result and the unresolved strong-backbone extensions are part of the same empirical account.

We make three contributions:

- We formulate bounded evidence completion as a separate layer over a dense ranking and specify a geometry-based, facet-conditioned implementation with protected insertion.
- We report repeated compact-backbone improvements and component, generator, and strong-backbone comparisons, retaining their distinct data boundaries and historical evaluation definitions.
- We explain what these comparisons identify through evidence-turnover analysis and mathematical clarification of granular-ball compactness, seed-relative coverage, and the limits of placement attribution.

2 Related Work

Retrieving connected evidence. Dense passage retrieval learns representations for relevance ranking (Karpukhin et al., 2020). Multi-hop systems can instead retrieve iteratively: MDR conditions later retrieval on earlier evidence (Xiong et al., 2021), whereas IRCOT interleaves retrieval and reasoning (Trivedi et al., 2023). Our intervention is a single completion of a fixed candidate ranking. It does not search an external corpus or rewrite queries using generated reasoning.

Structured representations. RAPTOR uses recursively summarized retrieval trees (Sarathi et al., 2024); HippoRAG uses knowledge-graph structure and propagation (Gutiérrez et al., 2024). HyperGraphRAG represents n-ary knowledge through hypergraphs (Luo et al., 2025). Our facet hyperedges instead relate query-local sentence groups. They are lexical selection relations, not extracted factual n-ary assertions. These systems motivate structure-aware retrieval, but their published results are not direct baselines under our candidates, generator, and budget.

Diversity, coverage, and context use. MMR balances relevance and redundancy (Carbonell and Goldstein, 1998). Coverage and diversity can also be represented by explicit submodular objectives (Lin and Bilmes, 2011). Such methods are important alternative explanations for a completion gain; the historical study did not run budget-matched

versions of them. We therefore compare the facet selector only with its actual centroid-only control. Long-context position effects (Liu et al., 2024) motivate protection, but do not establish that a particular protected-prefix length is optimal.

3 Method

3.1 Evidence Budget and Local Organization

For question q , let U_q be its benchmark-supplied candidate sentence units. The retriever embeds the query and each title-sentence pair, normalizes the embeddings, and scores

$$s(q, u) = \hat{e}_q^\top \hat{e}_u. \quad (1)$$

Descending score and ascending unit identity determine the dense order D_q . Every arm returns $K_q = \min(20, |U_q|)$ unique units. The compact variant uses the same MiniLM space for ranking and organization.

For a nonempty group B , define

$$c_B = \text{norm} \left(\sum_{u \in B} \hat{e}_u \right), \quad (2)$$

$$r_B = \max_{u \in B} (1 - \hat{e}_u^\top c_B).$$

The root contains all candidates, not only Dense Top-20. The compact implementation attempts a split when depth is below six, $|B| \geq 4$, and $|B| > 3$ or $r_B > 0.78$. The radius condition is redundant under these size settings. A farthest-from-center seed and its least-similar partner partition the group by cosine similarity. Both children must contain at least two units; otherwise the group is retained. Consequently, three is not a strict maximum leaf size. Geometry determines assignments, but radius does not independently trigger splits in this historical configuration.

The resulting groups are termed granular balls, following the broader idea of granular representations (Xia et al., 2019). This terminology does not assert that these groups outperform ordinary clustering.

3.2 Static Facet-Conditioned Selection

Let T_q be the content terms of the query under the fixed ASCII-alphanumeric tokenizer and stop-word list. A ball’s facets are the intersection of T_q with the union of terms in its titles and sentences. The two balls with highest query-centroid score form the seed set. An expansion hyperedge relates these

seeds to a non-seed ball contributing at least one additional query term.

For candidate B , let n_B and t_B denote seed-relative new and total facet counts; $m = |T_q|$, $b_B = \hat{e}_q^\top c_B$, a_B is maximum seed-centroid similarity, and ρ_B is the fraction of its facets already in the seeds. The historical compact score is

$$h_B = .45 \frac{n_B}{\max(m, 1)} + .20 \frac{t_B}{\max(m, 1)} + .20b_B + .15(1 - a_B) - .20\rho_B - .05|B|/8. \quad (3)$$

Eligibility additionally requires $|B| \leq 8$, $b_B \geq .12$ or $a_B \geq .05$, at most six units per new term, $\rho_B \leq .90$, and $h_B \geq .10$. At most two expansion balls survive descending h_B and stable edge-identity ordering. Units inside them retain their dense-score order.

All scores are computed once relative to the seeds. Accepting one ball does not change another’s score. Two accepted balls may therefore contribute the same new term. Nor does selecting a ball imply that every facet in its union occurs in the selected sentence or survives prompt truncation.

3.3 Protected Insertion

The compact policy preserves the first $\min(10, K_q)$ dense units, traverses the candidate order, and inserts at most four units absent from that prefix with score at least 0.1957079917192459. This historical development-calibrated 25th-percentile floor gives the policy its internal name, static-q25; it is not re-estimated per confirmation query. Remaining dense units are appended in order, with stable deduplication and truncation to K_q .

A placed unit can already belong to Dense ranks 11–20. We distinguish such a promotion from a genuinely new unit outside Dense Top-20. Equal K_q also does not imply equal token count: all methods share a 4,096-token input cap, while actual evidence length varies.

3.4 Two Extensions around BGE

The cross-space *sidecar* keeps BGE’s primary ranking and uses the frozen MiniLM structure to propose candidates outside BGE Top-20. Its protected and unprotected arms use the same insertion candidates but place them after the prefix or at the front, respectively.

The *native* variant reconstructs groups and relative scores in BGE space. Development selected

one of 16 configurations, with leaf target six, four insertion slots, unit-score percentile floor .50, and broad relative facet gates. The split depth, balance gate, score normalization, and candidate score differ from the compact method (Appendix A). This is a related implementation, not an encoder-only substitution. None of the results below comes from the later dynamic-selection prototypes.

4 Experimental Design

4.1 Questions and Comparisons

We use HotpotQA training distractor candidates and the answerable MuSiQue training candidate sets (Yang et al., 2018; Trivedi et al., 2022). The compact evaluations contain 1,000 HotpotQA questions, 3,000 MuSiQue questions, and a separate joint component set of 1,000 and 1,500 questions. The sidecar and native confirmation studies each use another 1,000/1,500 set. Their frozen identity manifests record exclusion of the preceding evaluation boundaries. These are project-held-out subsets of training releases, not official hidden test submissions.

The joint component study includes MiniLM Dense, BM25, a Dense–BM25 hybrid, BGE large-v1.5, FULL, NoFacet, and NoProtection. NoFacet retains compact ball geometry but uses centroid-only expansion. NoProtection changes the placement policy. Neither is a flat-unit or budget-matched MMR control. Native development uses 500 HotpotQA and 750 MuSiQue questions, separate from confirmation. Its small positive development difference selects a configuration; it is not pooled with confirmation.

4.2 Generation and Metrics

The principal generator is Qwen2.5-1.5B-Instruct FP16 on an RTX 4060 Laptop 8GB. A fixed short-answer prompt requests an answer from the evidence or UNKNOWN. Generation is greedy, batch size one, with at most 32 new tokens and a 4,096-token input cap. A transfer study uses the same initial rankings with Gemma 4 E2B in the official mobile-QAT deployment format. That comparison concerns these deployment configurations, not pure model architecture or device-independent efficiency.

We report mean answer F1 and exact match (EM). Complete recall (CR@20) records whether all annotated supporting evidence is covered; evidence recall (ER@20) records the covered frac-

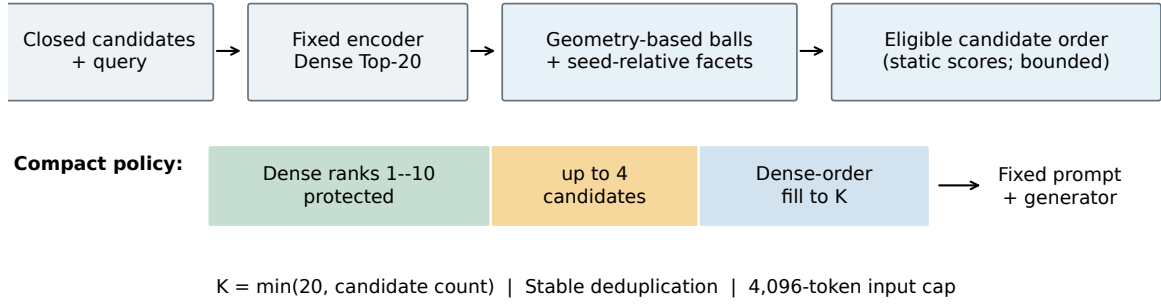


Figure 1: The historical compact pipeline. Candidate groups cover the full closed candidate set. Selection is static relative to the seeds; the protected prefix, candidate placement, dense-order fill, and subsequent token cap are separate operations. No answer label enters retrieval or prompting.

tion. HotpotQA has sentence-level support labels, whereas MuSiQue uses supporting paragraphs. MuSiQue coverage therefore must not be described as sentence-chain recall.

The initial HotpotQA and generator-transfer scorers include the special-answer F1 rule for yes/no/noanswer; the MuSiQue scorer uses answer aliases. The joint component, sidecar, and native studies use their frozen alias-max normalized token-overlap implementation without HotpotQA’s special-answer exception. We label these *legacy scoring* results and retain them unchanged. The impact on actual predictions has not been quantified, so these rows are not asserted to reproduce official HotpotQA leaderboard metrics.

4.3 Estimates and Uncertainty

For paired per-query scores, the target is the mean within-query difference. Joint studies average the two dataset means with equal weight. Their 10,000 bootstrap draws resample queries within each dataset, preserving method pairing; we reproduce the recorded percentile intervals rather than re-estimating them. The intervals describe the frozen query-sampling model, not uncertainty over model training or alternative development searches.

Statistical tests require a null distribution and suitable sampling assumptions (Dror et al., 2018). Some historical decision files call an uncentered bootstrap nonpositive-tail fraction a one-sided p-value and apply Holm adjustment. Its general null calibration was not established. We preserve the recorded decisions for provenance, but base the present description on effect estimates and intervals, without claiming a newly validated family-wise test. Intervals crossing zero are unresolved,

not equivalent. Shared source documents can induce dependencies not captured by query-level re-sampling.

5 Results

5.1 Does Completion Help Compact Retrieval?

Yes, within the tested compact configuration. Table 1 shows positive effects on the initial HotpotQA and MuSiQue sets and the separate component set. Their recorded 95% intervals exclude zero, with absolute F1 differences between .01140 and .01478. The initial HotpotQA lower bound is close to zero, so the magnitude and uncertainty are more informative than a binary success label.

Retrieval coverage also increases: CR@20 rises from .737 to .798 on HotpotQA and from .589 to .650 on MuSiQue. These are different support granularities, and neither change by itself measures answer improvement. The corresponding answer gains supply the end-to-end evidence.

5.2 What Do the Component Contrasts Identify?

On the joint compact set, FULL minus NoFacet is +0.01336 with interval [0.00341, 0.02343]. This favors the particular facet-conditioned selector over centroid-only expansion. It does not show that hyperedges are necessary, or that higher-order structure explains more than a simpler coverage heuristic. FULL minus NoProtection is +0.00354 [−0.00675, 0.01389], leaving the independent benefit of protection unresolved in that compact setting.

In the sidecar study, Protected minus Unprotected is +0.01122 [0.00129, 0.02104]. Protected

Evaluation set	Queries	Dense F1	Full F1	$\Delta F1$	95% interval
HotpotQA	1,000	.42150	.43628	+0.01478	[0.00020, 0.02988]
MuSiQue	3,000	.13595	.14735	+0.01140	[0.00450, 0.01835]
Joint component [†]	2,500	.30446	.31803	+0.01357	[0.00491, 0.02233]

Table 1: Compact MiniLM with Qwen. Scores are on the 0–1 scale. The joint row is dataset-equal-weighted, not a pooled average of all 2,500 questions. [†]Legacy scoring as defined in Section 4.2. All numbers reuse the frozen results.

insertion is better than the tested front insertion policy. Yet Protected minus BGE remains unresolved; avoiding part of an intervention’s damage is different from improving the untouched baseline. The native facet and placement contrasts also cross zero (Figure 3).

5.3 Does the Benefit Extend to Strong Retrieval?

The original compact FULL is below BGE on the same component set: -0.03998 $[-0.05393, -0.02621]$. BM25 and the hybrid are additional comparators, with FULL differences of -0.00755 $[-0.02259, 0.00773]$ and -0.00320 $[-0.01591, 0.00942]$, respectively. Neither interval resolves a difference.

The two BGE extensions address more specific alternatives. The sidecar adds candidates from the existing compact structure; its Protected-minus-BGE effect is -0.00256 $[-0.00998, 0.00458]$. Rebuilding the geometry natively in BGE yields -0.00305 $[-0.00688, 0.00063]$. Both point estimates are slightly negative, and both intervals cross zero. These findings do not establish complementarity with BGE, but they are not equivalence tests. Because the configurations and query sets differ, the three BGE comparisons are not a causal dose–response curve for retriever strength.

5.4 Does the Result Transfer to Another Generator?

With the fixed Gemma mobile-QAT configuration, the equal-weight FULL-minus-Dense effect is $+0.00516$ $[-0.00262, 0.01295]$. This result does not establish generator transfer. The rankings are reused from the initial studies, so this is an intervention on the tested generator deployment, not an additional independent retrieval sample. Differences in format and runtime prevent interpretation as a controlled architecture comparison between Qwen and Gemma.

6 Analysis and Implications

6.1 Adding Evidence Can Also Displace It

The post-decision sidecar audit counts 21 added and 29 displaced annotated supporting units in HotpotQA, and 59 added versus 50 displaced in MuSiQue. The corresponding native counts are 3/2 and 16/15. Thus, native completion has a net +1 supporting unit in each dataset while both answer-F1 point estimates are negative. These totals concern annotated evidence identities at each dataset’s support granularity, not complete reasoning chains or a causal effect per inserted unit.

A gain can fail to reach the answer because retained evidence is redundant, an essential fact is displaced, or the generator fails to use the context. The aggregate counts do not distinguish these mechanisms. They demonstrate why selection, final ranking, prompt-visible evidence, and answer quality should be measured separately.

6.2 What Geometry Does and Does Not Establish

For unit vectors x_i in a ball, compactness is

$$C(B) = \frac{\|\sum_{i \in B} x_i\|}{|B|}. \quad (4)$$

For any partition $B = \bigsqcup_j B_j$, the triangle inequality gives

$$\sum_j \frac{|B_j|}{|B|} C(B_j) \geq C(B). \quad (5)$$

This holds for arbitrary partitions. An increase in weighted child compactness therefore cannot alone validate a semantic split. The zero-sum scalar extends continuously to zero, although its normalized center is undefined without an implementation convention. These facts clarify the historical representation; they neither invalidate the observed compact gains nor independently explain them.

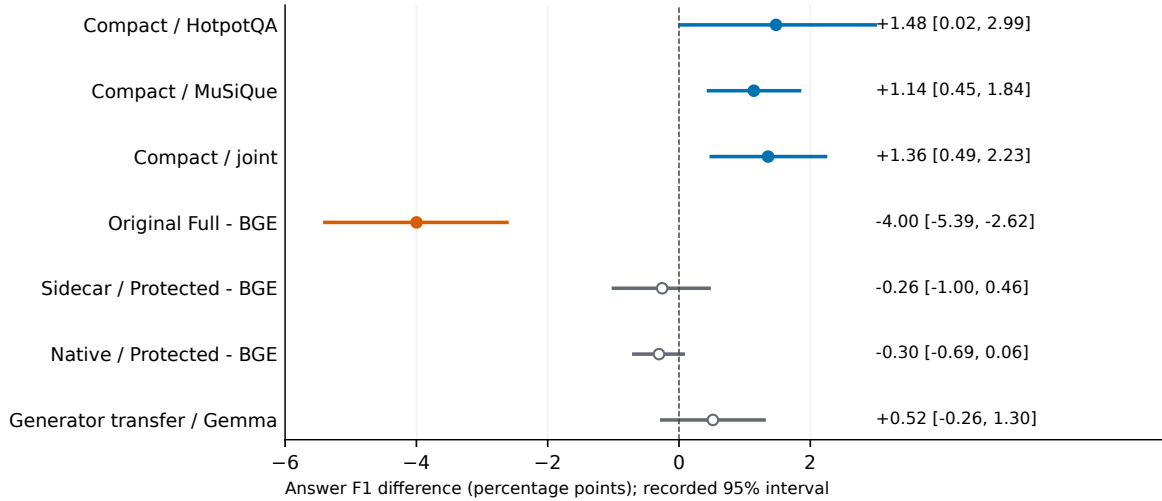


Figure 2: Paired answer-F1 effects in percentage points (one point is .01 on the score scale). Each interval is the recorded 95% percentile bootstrap interval for that row’s comparison; rows are not pooled. Open markers denote intervals crossing zero. Joint, original-versus-BGE, sidecar, and native rows use legacy scoring.

6.3 Evidence for a Layer, Not an Indispensable Component

The tested pipeline can improve a compact backbone without demonstrating that every named component is necessary. Ordinary clusters, MMR, and sentence-level coverage remain plausible alternatives. A decisive comparison would match candidate access, intervention budget, group count, and generator context. The absence of that comparison narrows the present contribution to a complete pipeline and specific component contrasts.

Similarly, the cross-space placement result does not prove that moving identical final evidence changes the answer. Final Top-20 membership and token truncation must also be controlled for that claim. A protected policy can help through preserving evidence, through order, or through both.

7 Limitations

All answer-quality results use closed benchmark candidate sets. They do not measure corpus reachability, ANN recall, full-wiki indexing, or open-domain noise. The positive compact evidence uses one retriever, one principal generator, one short-answer prompt, and Top-20. The additional generator and BGE extensions are unresolved rather than demonstrations of broad robustness.

The radius gate is inactive in the historical compact split rule, and the facet selector does not dynamically recompute coverage. A corrected or dynamic implementation is a different method and has no formal result in this paper. The native No-

Facet score changes multiple coefficients; its effect is not a pure component ablation. No fair ordinary-clustering or generic coverage comparison has been completed.

Legacy evaluator compatibility and historical tail-probability calibration limit the strength of statistical claims. We have not rescored predictions or replaced the frozen inferential procedure. Query-disjoint samples are not necessarily independent at the document or pretraining level. Dataset training releases may have appeared in generator pretraining. Post-decision mechanism summaries and representative cases do not identify causal mediation, and finite resource-based sample sizes are not a power guarantee.

8 Conclusion

Structured evidence completion improves the tested compact MiniLM pipeline on three closed-candidate multi-hop QA evaluations. A facet-conditioned selector also exceeds its specified compact centroid-only comparator. The original system remains below BGE, while cross-space and native BGE extensions do not establish an incremental answer-quality benefit. Protection helps relative to front insertion in one sidecar setting, but does not make its candidates useful relative to BGE. The resulting design lesson is to evaluate completion jointly with its backbone and displacement policy, using answer quality rather than structural coverage alone as the endpoint.

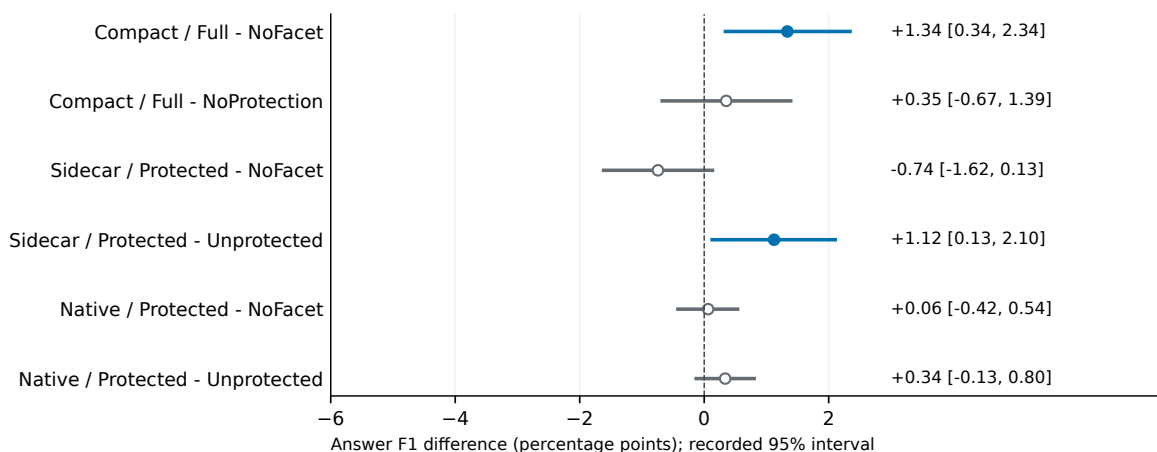


Figure 3: Component contrasts under legacy scoring. NoFacet is a specific historical comparator; native NoFacet also changes the candidate-score coefficients, so that row is a pipeline contrast rather than a pure removal of one coefficient. Placement comparisons retain insertion candidates but do not establish identity of the actual truncated prompt evidence.

Ethical and Reproducibility Considerations

This study uses existing benchmark releases and model artifacts; it introduces no new participant study. Negative and unresolved results are retained. Source code, configurations, derived predictions, and result manifests exist in the research repository. Third-party datasets and weights remain subject to their original terms. An anonymous distribution and author-approved licensing statement are required before submission. AI-assisted coding, analysis preparation, and writing were used; responsibility for the submitted work remains with the human authors.

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A Historical Implementation and Provenance

The compact method uses all-MiniLM-L6-v2; the strong ranking uses BAAI/bge-large-en-v1.5. Native C10 uses depth limit eight, minimum child fraction .20, and compactness-gain threshold 10^{-10} . Its two seed balls are ranked by query-score percentile, density percentile, then identity. Broad gates require query-score percentile at least .35 and density percentile at least .25. Eligible non-seed balls must contribute a seed-relative new facet; at most four balls supply candidates. Only units outside BGE Top-20 with unit-score percentile at least .50 are eligible. Candidate scoring is

$$h_u^{\text{native}} = .45p_u + .25p_{b(u)} + .15p_{\rho(u)} + .10p_{C(u)} + .05 \min(n_{b(u)}, 3)/3, \quad (6)$$

where the p terms are within-query empirical percentiles of unit relevance, ball relevance, density, and compactness. Native NoFacet replaces this by $.50p_u + .30p_{b(u)} + .10p_{\rho(u)} + .10p_{C(u)}$ and removes the new-facet eligibility requirement. The comparison therefore changes both gating and weights.

The paper’s scientific names map to the frozen repository records as follows: initial HotpotQA/MuSiQue to Stage4E/4F, generator transfer

to Stage4G, joint components to Stage4H, cross-space sidecar to Stage4I, and native confirmation to Stage5A. Aggregate summaries, implementation configurations, and the manuscript evidence ledger provide the exact artifact paths. The later Stage6 development outputs, engineering repairs, and synthetic v2 prototype are excluded from these empirical results.

Verification scope. Historical verification records are retained as records of their original checks. This manuscript revision cross-checks the reported aggregate values and plotting inputs; it does not rerun retrieval, generation, Gold evaluation, or bootstrap. Agreement of files or schemas cannot establish independent semantic correctness, and subsequent verifier repairs are not retroactive certification of the old experiments.

Complexity and measured cost. With embeddings available, recursive two-seed assignment costs $O(H|U_q|d)$, scoring and sorting cost $O(|U_q| \log |U_q| + |\mathcal{B}_q|d + |\mathcal{B}_q| \log |\mathcal{B}_q|)$, and working memory is $O(|U_q|d + |U_q| + |\mathcal{B}_q|)$, excluding tokenization and model inference. The native confirmation records 8.544 s retrieval reconstruction, 887.650 s BGE generation, 884.198 s protected-native generation, and 3.592 GiB peak GPU memory. These are transaction-specific measurements, not corpus-index costs or a controlled speed advantage.